

JUSTIN DUC

ML Systems & Kernel Optimization Researcher

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EXPERIENCE

ML Optimization Consultant

Audioshake

09/2025 – 04/2026 Remote (California)

- Engineered high-throughput inference pipelines across TensorRT 10.x, OpenAI Triton, Helion, torch.compile, and ONNX, targeting NVIDIA H100 and A100 architectures.
- Authored specialized Triton custom ops for audio source separation models, maximizing Tensor Core utilization to reduce VRAM memory bandwidth by an estimated 40% and cut kernel launch latency.
- Developed automated transpilation frameworks bridging custom Python @triton.jit operators directly to ONNX Runtime (OrtCustomOp) and TensorRT C++ plugins, increasing inference throughput by up to 2.5x compared to eager PyTorch execution.

Owner & Technical Founder

Eclat

01/2024 – Present Lyon, France

- Architected automated B2B 3D jewelry customization engine, designing Python scrapers interfacing directly with industrial Materialise manufacturing APIs.
- Deployed the entire full-stack application using Docker containers orchestrated for high availability, handling procedural Blender 3D rendering pipelines and achieving a 20% field conversion rate.
- Formulated financial pricing algorithms accounting for live precious metal spot prices, URSSAF tax rates (12.3%), and €50 Minimum Order Value (MOV) supplier cliffs.

Intern Teacher

EPITECH

09/2025 – 02/2026 Lyon, France

- Supervised, graded, and mentored 50+ computer science students on Machine Learning, Deep Learning architectures, and C++ systems programming.

AI Developer

Wanadev

02/2024 – 06/2024 Lyon, France

- Formulated mathematical game balancing simulations and AI agent decision models for a commercial autobattler game.

Teacher

ISEG

01/2024 – 03/2024 Lyon, France

- Taught relational database architecture (SQL), PHP, and web development fundamentals to undergraduate students.

Full-Stack Developer

SNB Technologies

10/2023 – 02/2024 Remote

- Developed administrative software and backend analytics pipelines for enterprise conversational AI chatbots.

RESEARCH & PROJECTS

Kernellens: Triton-to-C++ Compiler

Galiad Research

02/2026 – 10/2026 Lead Author

- Architected an open-source 4-phase compiler converting PyTorch @triton.jit kernels to ONNX Runtime & TensorRT C++ plugins via PTX extraction and CUDA Driver API.
- Implemented SymPy AST grid transpilation, 16-byte CUDA alignment guards, and Blackwell (sm_120a) PTX ISA version clamping, specifically optimizing Flash Attention execution on next-gen hardware.
- Achieved $3.4\times - 7.0\times$ speedups over eager PyTorch across LLaMA 3, Qwen 2.5, and Liger-Kernels with exact numerical parity ($MaxDiff = 0$).
- Published research paper (Zenodo DOI: 10.5281/zenodo.23194523) establishing the tool as a robust deployment artifact.

TreaT: Tree Attention for Retrieval

Tsinghua University

01/2025 – 06/2025 Lead Author

- Designed a novel binary memory tree attention replacing dense $O(N^2)$ comparisons with $O(\log N)$ logarithmic-time routing over hierarchical summaries.
- Formulated analytical routing error propagation models; trained via self-attention distillation with Reciprocal-Value Scoring.

Saute: Utterance Embeddings

Tsinghua University

01/2025 – 06/2025 Lead Author

- Developed a linear-attention Transformer architecture leveraging speaker-sensitive memory banks for efficient utterance-level context modeling.

EDUCATION

Study Abroad in AI & NLP

Tsinghua University

09/2024 – 07/2025 Beijing, China

Advanced research in Deep Learning, NLP, Algorithm Design, Big Data Intelligence, and Quantitative Finance.

Expert in Computer Science

EPITECH

2021 – 2026 Lyon, France

Master's level degree in Software Engineering, Architecture, and Systems.

AWARDS

ML Research Developer

POC

10/2022 – 06/2023 Lyon, France

- Led 6-month AI research initiatives on VQ-VAE generative style transfer and partnered with Effscience on neural network feature interpretability.

Full-Stack Developer

Visiativ PLM

06/2022 – 12/2022 Bron, France

- Refactored and ported a legacy enterprise Java desktop PLM application into a modern, high-performance Angular web application.



Pioneer Award 2024
Chengdu 80 Fintech Design Competition at Tsinghua University.

SKILLS

OpenAI Triton

NVIDIA TensorRT 10.x

CUDA C++

ONNX Runtime

Docker

PyTorch Compiler

SymPy AST

PTX ISA Assembly

C++17/20

Python

LLM Inference

LANGUAGES

French

English

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